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title: "Technical Specification: AI-Native DataOps Platform & Sentinel Engine"
author: "Manus AI"
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Technical Specification: AI-Native DataOps Platform & Sentinel Engine

1. Executive Summary

The transition from traditional data engineering to AI-Native engineering requires a fundamental shift in both architecture and operational philosophy. This document outlines the technical specification for building a modern, end-to-end analytical data platform coupled with an autonomous DataOps monitoring system.

The objective is to construct a system where data pipelines are orchestrated deterministically, while operational monitoring, root cause analysis, and remediation are handled by an autonomous Multi-Agent system. Furthermore, the platform exposes its analytical capabilities not just through static dashboards, but via a Model Context Protocol (MCP) server, enabling natural language querying and dynamic report generation through AI interfaces like Claude Desktop.

This architecture demonstrates the complete AI-Native stack: deterministic infrastructure (Dagster, dbt, DuckDB), intelligence exposure (FastAPI + MCP), and autonomous operations (CrewAI Sentinel Engine).

2. Use-Case: E-Commerce Analytics Backbone

The Business Context

A high-volume e-commerce company currently relies on a monolithic PostgreSQL database as its primary transactional system. This database handles all core operations: order processing, customer management, inventory tracking, and payment recording.

The Problem Statement: Why PostgreSQL Analytics is Fundamentally Broken

The company's current analytics infrastructure is built on a flawed architectural premise: using PostgreSQL (a transactional database) as a columnar analytics engine. This approach

was never designed for analytical workloads and is now causing cascading failures across the organization.

The analytics team is struggling with three critical pain points. First, analytical queries (e.g., "Top selling products by region over the last 30 days") are executed directly against the transactional PostgreSQL database, causing severe performance degradation during peak sales events. This directly impacts customer-facing operations, as heavy analytics queries lock resources needed for order processing. Second, the system suffers from fundamental data fragility. Schema evolution in the application layer—such as adding new product attributes or changing payment methods—frequently breaks downstream reporting, forcing emergency fixes and manual data reconciliation. Finally, the operational overhead is unsustainable. Data engineers spend the majority of their time debugging pipeline failures, tracking down late-arriving data, and writing manual post-mortem reports instead of building new analytical capabilities. The company is essentially paying premium engineering salaries to maintain a broken system rather than innovate.

The AI-Native Solution: A Purpose-Built Analytical Backbone

To resolve these issues, we will construct a fundamentally different architecture: a **purpose-built analytical backbone** that separates transactional and analytical concerns entirely. This new system extracts data from the transactional PostgreSQL database, transforms it using modern data engineering practices (dbt + Medallion architecture), and materializes it in DuckDB—a high-performance analytical engine designed specifically for this workload.

Crucially, this system will not rely on manual operational monitoring or static dashboards. Instead, it will be supervised by the **Sentinel Engine**, an autonomous CrewAI-based multi-agent system that continuously monitors the pipeline, detects anomalies, diagnoses root causes, and proposes fixes without human intervention. This transforms data engineering from a reactive, debugging-heavy discipline into a proactive, autonomous operation.

Finally, the analytical data will be exposed via a **Model Context Protocol (MCP) server**, enabling stakeholders to interact with the platform using natural language through interfaces like Claude Desktop. This democratizes data access, allowing business users to ask analytical questions directly without requiring engineers to build and maintain static dashboards.

3. Validation Strategy: Synthetic Data Generator

Before deploying this new analytical backbone to production, the system must be rigorously validated against realistic failure scenarios. To accomplish this, we will build a **Synthetic Data Generator**—a permanent testing harness that simulates both normal e-commerce traffic and the exact failure modes observed in the current production environment.

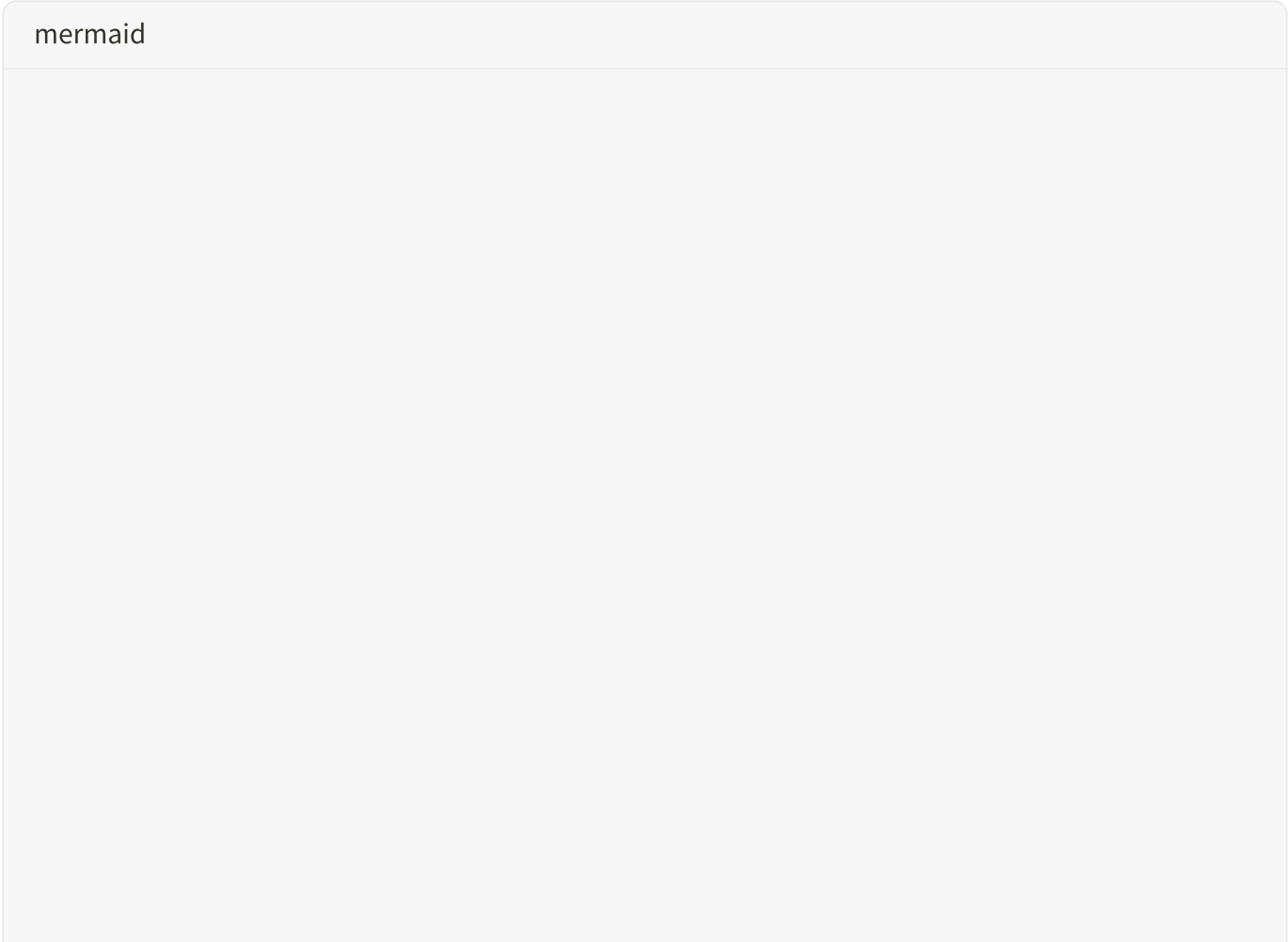
The Synthetic Data Generator serves two critical purposes. First, it enables **smoke testing**: verifying that the entire pipeline (Dagster → dbt → DuckDB → MCP) functions correctly with realistic data volumes and schemas. Second, it enables **stress testing**: injecting deliberate failures (schema changes, data quality issues, late-arriving data, upstream failures) to validate that the Sentinel Engine can detect, diagnose, and remediate these issues autonomously.

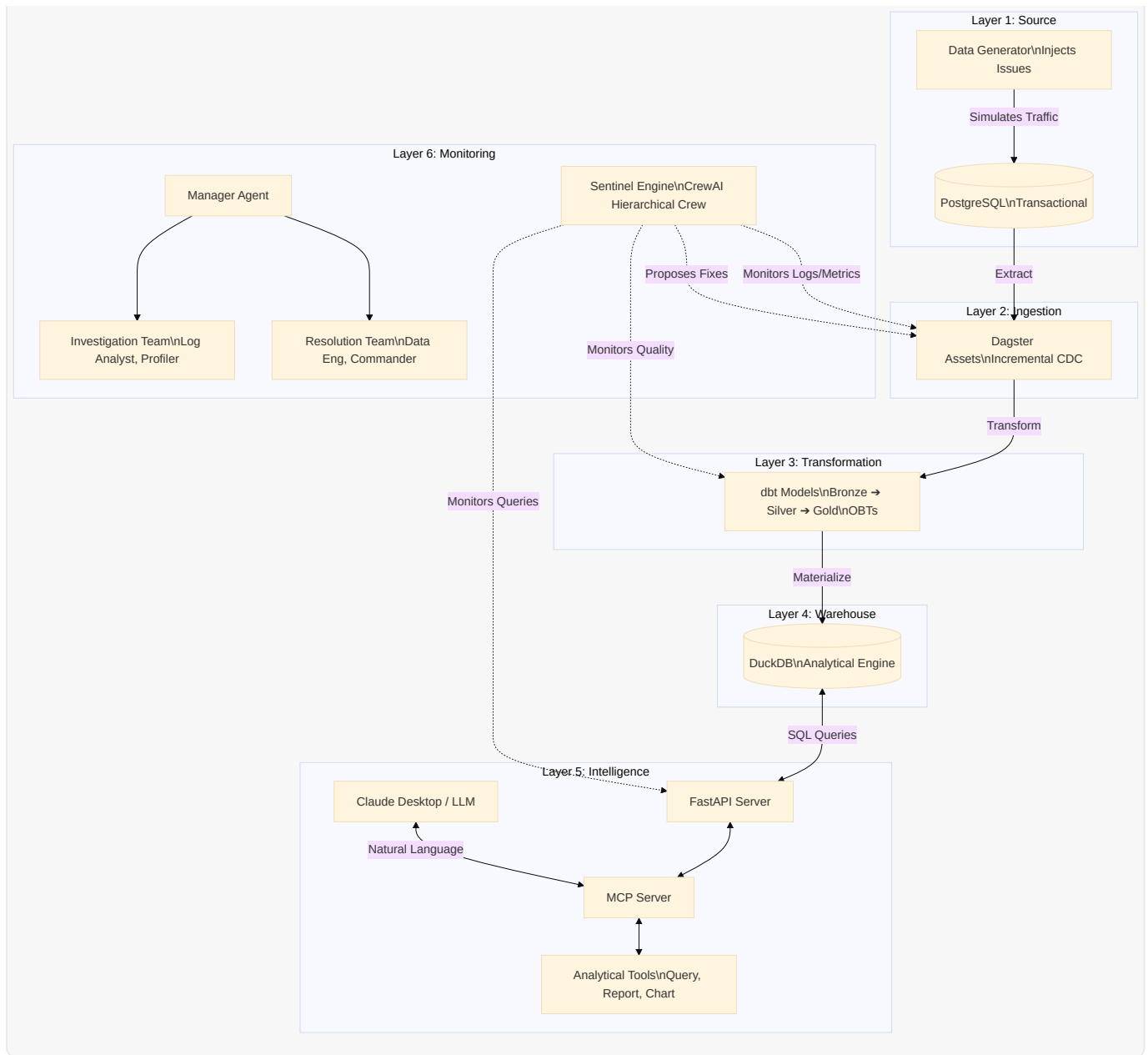
This validation approach ensures that when the system is promoted to production, it has been battle-tested against the exact chaos it will encounter in the real world. The Data Generator becomes a permanent fixture—teams can inject new failure modes anytime to continuously validate system resilience. This is not a demo tool; it is a **production validation harness** that proves the system is ready for enterprise workloads.

4. Architecture: The 6-Layer AI-Native Platform

The platform is divided into two distinct logical systems: **The Analytical Platform** (Layers 1-5) which handles deterministic data movement and exposure, and **The Autonomous DataOps Engine** (Layer 6) which handles probabilistic monitoring and remediation.

High-Level Architecture Diagram





Layer Breakdown

Layer 1 (Source) consists of a PostgreSQL database acting as the primary transactional system. This environment is actively populated by a Python Data Generator that simulates both normal e-commerce traffic and realistic failure modes to stress-test the pipeline.

Layer 2 (Ingestion) utilizes Dagster to orchestrate the incremental extraction of data from the PostgreSQL source. Dagster ensures robust management of data lineage and execution dependencies.

Layer 3 (Transformation) employs dbt to apply software engineering best practices to data modeling. It transforms the raw ingested data into a structured Medallion architecture (Bronze, Silver, Gold), culminating in the generation of One-Big-Tables (OBTs) optimized for analytics.

Layer 4 (Warehouse) leverages DuckDB as the high-performance, embedded analytical engine, providing fast query execution without the overhead of a distributed warehouse.

Layer 5 (Intelligence) bridges the gap between data and users via a FastAPI server implementing the Model Context Protocol (MCP). This layer exposes DuckDB's analytical capabilities as deterministic tools accessible to any LLM interface, such as Claude Desktop.

Layer 6 (Monitoring) introduces the Sentinel Engine, a CrewAI-based autonomous team. This system continuously watches the entire stack, asynchronously diagnosing failures injected by the Data Generator and proposing actionable remediations.

5. The Analytical Platform (Layers 1-5)

This section specifies the deterministic components of the platform, built to extract, transform, and serve data via natural language interfaces.

5.1 Data Source & Ingestion (Dagster)

The foundation of the analytical platform begins with a PostgreSQL instance containing core e-commerce tables such as `orders`, `customers`, `products`, and `payments`. To simulate a live environment, a Python script (`data_generator.py`) acts as the generator, continuously inserting new records while periodically injecting anomalies like negative prices, missing customer IDs, or duplicate orders. The ingestion process is managed by Dagster Software-Defined Assets (SDAs), which perform reliable, incremental extraction of this data from PostgreSQL directly into raw DuckDB tables.

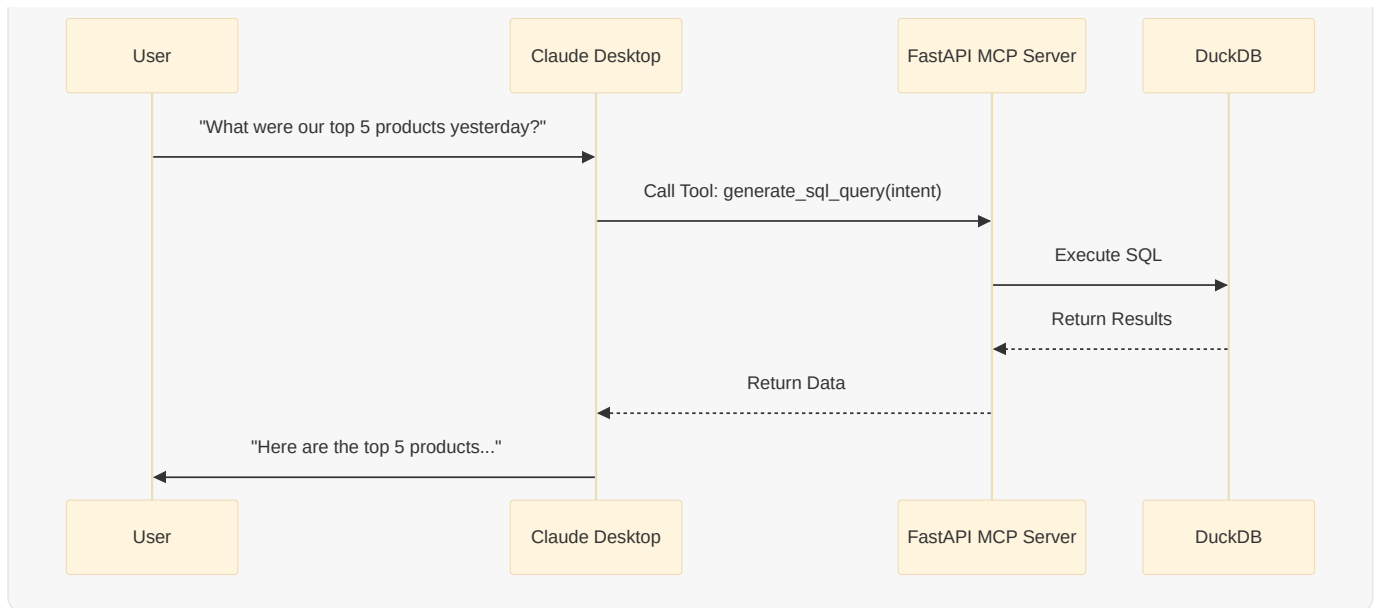
5.2 Transformation (dbt + DuckDB)

For the transformation layer, DuckDB serves as the execution engine, chosen for its exceptional high performance on analytical workloads without the complexity and overhead of a distributed warehouse. The transformation architecture strictly follows the Medallion pattern implemented via dbt. It begins with `bronze_` models acting as raw data mirrors, progresses to `silver_` models containing cleansed, deduplicated, and strongly typed data, and culminates in `gold_` models that provide business-level aggregates and One-Big-Tables (OBTs) highly optimized for querying.

5.3 Intelligence Exposure (FastAPI + MCP)

Instead of relying solely on static BI tools, the platform exposes its analytical power directly to AI agents.

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MCP Tools Specification:

The MCP server exposes three primary tools. The `get_schema_info` tool returns the schema of the `gold_` tables, allowing the LLM to fully understand the available data structure. The `execute_analytical_query` tool accepts a natural language intent, translates it into optimized DuckDB SQL, executes the query, and returns the deterministic result set. Finally, the `generate_report` tool accepts parameters for standard reporting (e.g., "Daily Sales"), executes a series of predefined queries, and formats the output for immediate consumption.

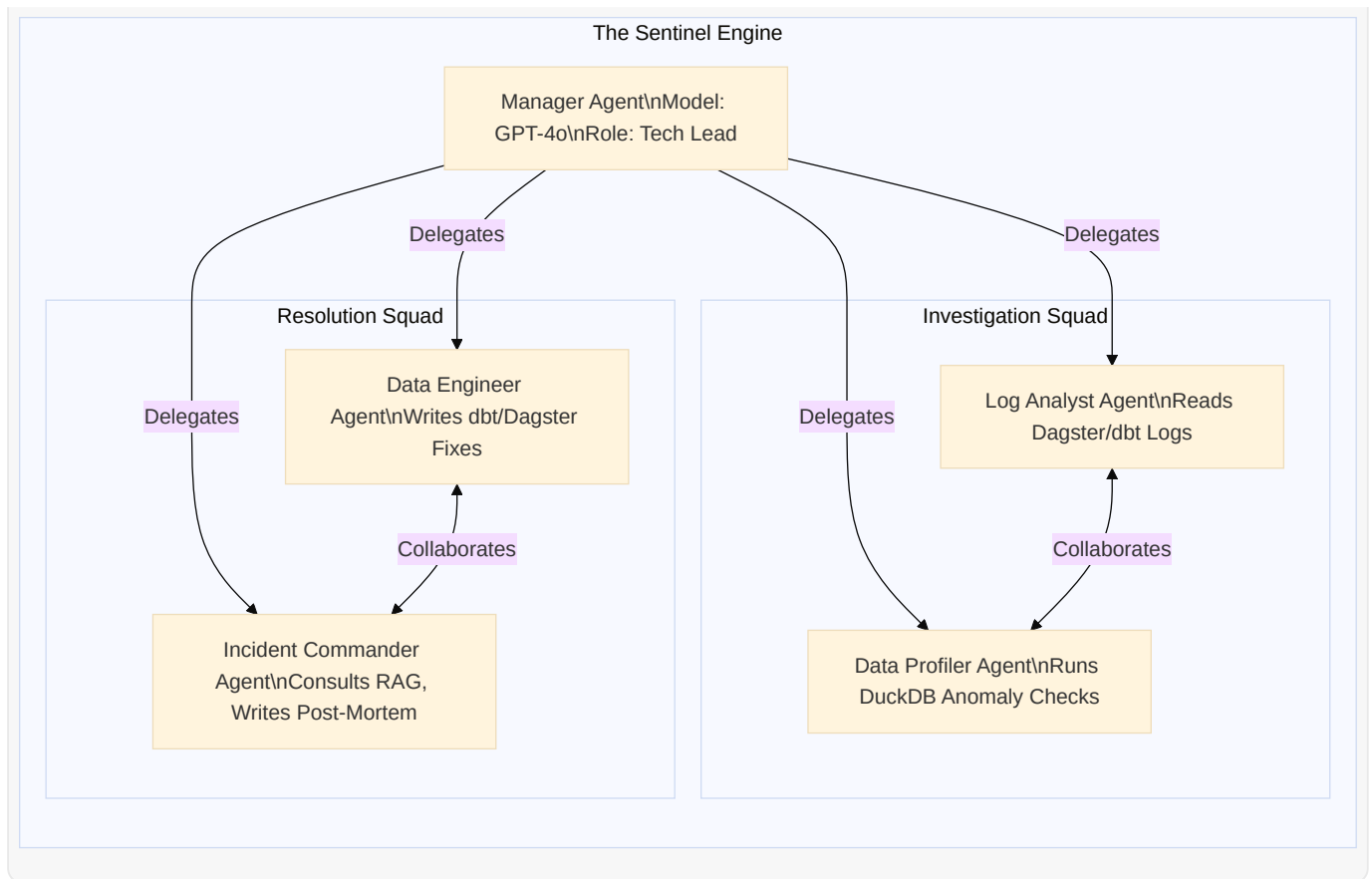
6. The Autonomous DataOps Engine: The Sentinel (Layer 6)

The Sentinel Engine is a separate, autonomous system built using CrewAI. It operates asynchronously, monitoring the Analytical Platform and responding to failures injected by the Data Generator.

6.1 Architecture & Design

The Sentinel uses a **Hierarchical Process** to simulate a real engineering team.

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6.2 AgentSpec Definitions

The entire Sentinel Engine will be generated using the AgentSpec framework. Below is the specification that an AI-Native Engineer would pass to an agent to build the Crew.

Agent: Manager (Tech Lead) Operating as the DataOps Tech Lead, this agent's goal is to coordinate the investigation of pipeline failures, ensure root causes are accurately identified, and validate proposed fixes before approving final post-mortems. It utilizes tools such as `AssignTask` and `ReviewOutput` to manage the team.

Agent: Log Analyst Functioning as the Orchestration Log Specialist, this agent parses Dagster and dbt logs to pinpoint the exact execution node and error trace responsible for a pipeline failure. Its primary tools include `ReadDagsterLogs` and `ReadDbtRunResults`.

Agent: Data Profiler Acting as the Data Quality Specialist, the Profiler queries DuckDB to identify statistical anomalies, schema drifts, or missing constraints that correlate with the reported pipeline failures, using tools like `QueryDuckDB` and `ProfileTable`.

Agent: Data Engineer Taking on the role of the Analytics Engineer, this agent is responsible for writing the specific Python (Dagster) or SQL (dbt) code required to resolve the root cause identified by the Investigation Squad. It leverages the `WriteCodePatch` and `ValidateDbtModel` tools.

Agent: Incident Commander Serving as the Incident Response Lead, this agent searches the company's historical incident RAG database for similar past issues and authors a

standardized, blameless post-mortem report. It relies on the `QueryHistoricalRAG` and `WriteMarkdownReport` tools.

6.3 The Remediation Loop

The autonomous remediation process follows a strict sequence. It begins with **Detection**, where the Data Generator injects a deliberate schema change, such as renaming `user_id` to `customer_id` in the PostgreSQL source. This immediately causes a **Failure** in the downstream Dagster or dbt pipeline. This failure acts as a **Trigger**, firing a webhook that alerts the Sentinel Manager.

The Manager then initiates the **Investigation** phase, assigning the Log Analyst to locate the broken dbt model in the execution logs, while the Data Profiler confirms the missing column directly within DuckDB. Finally, the **Resolution** phase occurs when the Manager assigns the Data Engineer to author the necessary dbt migration to fix the code, and tasks the Incident Commander with documenting the entire event in a comprehensive post-mortem report.

7. Infrastructure Deployment: Local & Cloud Native

The platform is designed to run entirely locally for development and demonstration, with a clear, low-friction migration path to cloud-native managed services.

7.1 Local Development (docker-compose)

For development and demonstration purposes, the entire stack runs locally orchestrated by a unified `docker-compose.yml` file. This environment includes a PostgreSQL container running the transactional database, alongside a Data Generator container executing the Python script that injects data and anomalies. A dedicated Dagster container hosts both the daemon and the Dagit UI for orchestration management. DuckDB operates as an embedded engine within the Dagster and dbt processes, while a separate FastAPI/MCP container securely exposes the DuckDB file to external LLM interfaces.

7.2 Cloud Native Scaling Path

When transitioning to a production environment, the core architecture remains completely identical, but the underlying infrastructure seamlessly shifts to managed cloud services. The local DuckDB file is simply replaced by a MotherDuck connection string, requiring absolutely no SQL rewrites. The local Dagster instance migrates to Dagster Cloud for serverless execution of the orchestration layer. The FastAPI application is deployed as a serverless container using services like Google Cloud Run or AWS Lambda. Finally, the Sentinel Engine's CrewAI process is transitioned to run as a continuous background job or as a serverless function triggered by cloud-native webhooks.

8. Cost Analysis & ROI

8.1 Budget Implications

Running the platform in a local environment incurs zero direct costs, as it leverages open-source tools and local compute resources. In a cloud environment, the costs are highly optimized and predictable. MotherDuck utilizes compute-based pricing, which is exceptionally efficient for targeted analytical queries. Dagster Cloud operates on a serverless tier billed by execution minutes. The primary variable cost is the LLM API usage, utilizing GPT-4o for the Sentinel Manager and GPT-4o-mini for the specialized agents, which is estimated at under \$50 per month for typical monitoring workloads.

8.2 Return on Investment (ROI)

The traditional data engineering approach requires human engineers to manually monitor logs, write SQL fixes, and document incidents—a highly inefficient use of expensive talent. The AI-Native approach shifts this operational burden entirely to the Sentinel Engine. This results in a drastic reduction of Mean Time to Resolution (MTTR) from hours to mere minutes. Consequently, engineering velocity increases significantly, as data engineers can focus on building new models and features rather than debugging existing pipelines. Furthermore, the MCP layer drives data democratization, allowing business users to query data naturally via Claude Desktop without requiring engineers to build and maintain static dashboards.

9. Success Metrics

The implementation of this platform will be evaluated against four primary success criteria. First, the Analytical Platform must successfully synchronize transactional data from PostgreSQL to DuckDB and execute transformations via dbt without errors. Second, the intelligence layer must allow Claude Desktop, connecting via MCP, to successfully query the DuckDB warehouse using natural language and return accurate results. Third, the Data Generator must successfully and predictably inject defined failure modes, such as schema drift, into the source system. Finally, the Sentinel Engine must autonomously detect the injected failure, accurately diagnose the root cause within the logs, and output a correct remediation plan alongside a comprehensive post-mortem report, all without human intervention.

10. Team Requirements

Building and maintaining this AI-Native DataOps Platform requires a modern skill set. The team needs an AI-Native Data Engineer proficient in modern data stack tools including Dagster, dbt, and DuckDB. Additionally, an Agentic Engineer is required, possessing the capability to write precise AgentSpecs, configure CrewAI hierarchies, and optimize LLM prompts for deterministic outputs. Finally, a Backend Engineer is needed to maintain the FastAPI MCP server and ensure the secure exposure of analytical tools. Notably, in a mature AI-Native paradigm, these distinct roles can often be seamlessly fulfilled by a single engineer adept at orchestrating specialized AI agents.